Course notes

Area and antiderivative

For f continuous and positive on $[a, b]$, $a \leq b$, the integral $\int_a^b f(x)\,\mathrm{d}x$ is the area under the curve, above the axis, between $x = a$ and $x = b$, expressed in units of area.

Fundamental theorem

If f is continuous on an interval I and $a \in I$, then $F_a(x) = \int_a^x f(t)\,\mathrm{d}t$ is the antiderivative of f that is zero at a. For every antiderivative F,

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Every function continuous on an interval has antiderivatives. An integral is a number; an antiderivative is a function.

Proof in the positive increasing case. For $h > 0$, the rectangles give $hf(x) \leq F_a(x+h) - F_a(x) \leq hf(x+h)$. After division by h and taking the limit, the rate of change tends to $f(x)$. For $h < 0$, monotonicity gives $f(x+h) \leq f(t) \leq f(x)$ on $[x+h, x]$. By integrating,

$$(-h)f(x+h) \leq F_a(x) - F_a(x+h) \leq (-h)f(x).$$

Dividing by $-h > 0$, we get

$$f(x+h) \leq \frac{F_a(x+h) - F_a(x)}{h} \leq f(x).$$

Both bounds tend to $f(x)$. By continuity, $F_a' = f$. Any other antiderivative is $F_a + C$, hence the formula for evaluating at the bounds.

Properties

For continuous functions and bounds in the same interval:

$$\int_a^a f = 0, \quad \int_b^a f = -\int_a^b f, \quad \int_a^c f = \int_a^b f + \int_b^c f, \quad \int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g.$$

If $a \leq b$ and $f \leq g$, then $\int_a^b f \leq \int_a^b g$. In particular, if $m \leq f \leq M$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$. For a function of any sign, the integral is defined using antiderivatives. The geometric area between the curve and the axis is $\int_a^b |f|$, obtained by splitting at the sign changes. Between two curves, it equals $\int_a^b |f - g|$.

Example. $\int_{-1}^{1} x \, \mathrm{d}x = 0$, but the area between the line and the axis on this interval is 1.

Mean value and integration by parts

For $a < b$, the mean value is $\mu = \frac{1}{b-a}\int_a^b f(x) \, \mathrm{d}x$. The rectangle of height μ and width $b-a$ has the same algebraic area.

Integration by parts

If u, v are of class $\mathcal{C}^1$ on $[a, b]$,

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

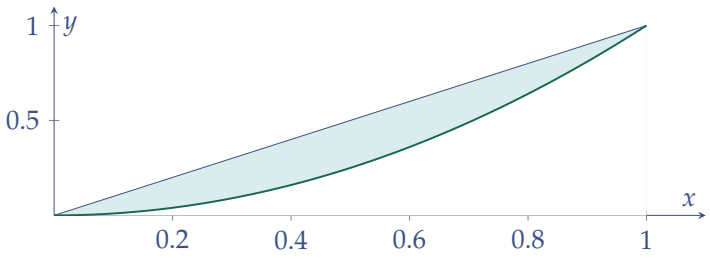
Proof. Integrate $(uv)' = u'v + uv'$ then isolate the required integral. The bracket is indispensable.

Example. With $u = x$ and $v' = e^x$, $\int_0^1 xe^x \, \mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \, \mathrm{d}x = e - (e-1) = 1$.

Rectangles and sequences of integrals

For f continuous and increasing on $[a, b]$, with $a < b$ and $n \geq 1$ and $h = \frac{b-a}{n}$, the sums $h\sum_{k=0}^{n-1} f(a+kh)$ & $h\sum_{k=1}^{n} f(a+kh)$ bound the integral. Their difference is $h(f(b) - f(a))$. For $I_n = \int_0^1 x^n f(x) \, \mathrm{d}x$, start with the sign and comparisons of the integrand. If $0 \leq f \leq M$, then $0 \leq I_n \leq \frac{M}{n+1}$; the squeeze theorem gives $I_n \to 0$. Integration by parts can also give a relation between I_n and I_{n+1}.

Convention for index zero. In $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, the expression x^0 denotes the constant function 1, including at the bound 0. Thus $I_0 = \int_0^1 f(x)\,\mathrm{d}x$; no indeterminate form needs to be resolved.



The area between $y = x$ and $y = x^2$ on $[0,1]$ is $\frac{1}{6}$.

Linking area, antiderivative and approximation

8.1 Two areas, one integral

First use: S59 and S61.

Worked example · A change of sign

On $[0;3]$, $f(x) = x - 1$ is negative before 1, positive after. The integral equals $[x^2/2 - x]_0^3 = 3/2$. The geometric area is $-\int_0^1 f + \int_1^3 f = 1/2 + 2 = 5/2$. The integral involves cancellation between two signed contributions.

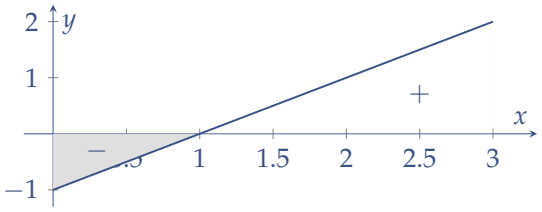


Figure 8.1. For a geometric area, the two regions add up; for the integral, their signs come into play.

A1 · Immediate application For $g(x) = 2x - 2$ on $[0;3]$, calculate the integral and the area between the curve and the axis. How does this relate to the values in the example?

8.2 Why the area gives an antiderivative

First use: S61.

Property · Fundamental theorem

For f continuous on an interval I and $a \in I$, $A(x) = \int_a^x f(t)\,\mathrm{d}t$ is an antiderivative of f. So, for any antiderivative F, $\int_a^b f = F(b) - F(a)$. The general case is accepted without proof; the positive increasing case is proved below.

Proof. Suppose f is continuous, positive and increasing. For h>0, the comparison with two rectangles gives $f(x) \le [A(x+h) - A(x)]/h \le f(x+h)$. For h<0, it gives $f(x+h) \le [A(x+h) - A(x)]/h \le f(x)$. By continuity and squeezing, the difference quotient tends to f(x); thus $A' = f$. Any other antiderivative F differs from A by a constant; since A(a)=0, $A(b) = F(b) - F(a)$.□

Worked example · Recovering a derivative

For $A(x) = \int_0^x 2t\,\mathrm{d}t$, the area is x^2 when x is positive. The antiderivative formula also gives $A(x) = x^2$ for negative x, with the orientation of the integral. In both cases $A'(x) = 2x$.

A2 · Partial guidance On $[0;+\infty[$, set $A(x) = \int_0^x e^{-t}\,\mathrm{d}t$. Give A(x), A'(x), the direction of variation and the limit. Interpret the limit as an area.

8.3 An equivalent height, with its units

First use: S63.

Definition · Mean value

For $a < b$, the mean value of f on $[a;b]$ is $m = \frac{1}{b-a}\int_a^b f$. If f is positive, the rectangle of width b-a and height m has the same area as the region under the curve.

Worked example · Flow rate and volume

A flow rate $d(t) = 2 + t$ L/min for three minutes delivers $\int_0^3 (2+t)\,\mathrm{d}t = 10{,}5$ L. Its mean flow rate is 10.5/3 $= 3.5$ L/min. The volume and the mean flow rate do not have the same unit.

A3 · Immediate application Calculate the mean of x^2 on $[0;1]$. Draw the equivalent rectangle. Explain why this mean value is not the mean of the two endpoint values.

8.4 Draw and check the bounds

First use: S63 and S66.

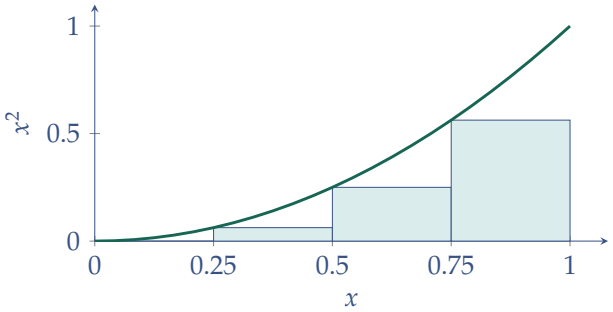


Figure 8.2. Left-hand rectangles. Complete the drawing with the right-hand rectangles and the line at height 1/3.

Worked example · Two intervals

For x^2 on [0;1], two left-hand rectangles give $L_2 = \frac{1}{2}(0^2 + (1/2)^2) = 1/8$; the right-hand ones give $U_2 = \frac{1}{2}((1/2)^2 + 1^2) = 5/8$. The integral 1/3 does lie between these bounds.

A4 · Investigate For $f(x) = x^2$ on $[0;1]$, divide into four equal intervals. Calculate the left and right sums; justify their positions relative to the integral. Then show that, for n intervals, their difference is 1/n.

A first step: In the difference of the two sums, the interior values cancel.

8.5 Choosing what to differentiate

First use: S66.

Property · Integration by parts

For u, v of class $\mathcal{C}^1$ on $[a;b]$, $\int_a^b u'v = [uv]_a^b - \int_a^b uv'$. The formula comes from integrating $(uv)' = u'v + uv'$.

Worked example · Two successive integrations by parts

For $I = \int_0^1 x^2 e^x \,\mathrm{d}x$, differentiating a polynomial lowers its degree. First $I = [x^2 e^x]_0^1 - 2\int_0^1 xe^x \,\mathrm{d}x$. Then $\int_0^1 xe^x \,\mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \,\mathrm{d}x = e - (e-1) = 1$. Hence I=e-2. Writing the bounds at each step avoids forgetting a constant.

A5 · Choice of strategy Calculate $\int_1^e \ln x \,\mathrm{d}x$. Justify your choice in the integration by parts and check the result by differentiating an antiderivative.

8.6 Bound before looking for a formula

First use: S66.

Worked example · Bounds without an explicit antiderivative

Let $J_n = \int_0^1 x^n/(1+x)\,\mathrm{d}x$. Since $1/2 \le 1/(1+x) \le 1$, $1/[2(n+1)] \le J_n \le 1/(n+1)$. Both bounds tend to 0, so $J_n \to 0$, without calculating Jn for each n.

A6 · Synthesis For $n \in \mathbb{N}$, $I_n = \int_0^1 x^n e^{-x} \,\mathrm{d}x$. Show that (I_n) is positive and decreasing. Establish $e^{-1}/(n+1) \le I_n \le 1/(n+1)$, then its limit. Does integration by parts give a relation between I_n and I_{n+1} ?

A7 · Refute Give a non-zero continuous function on $[-1;1]$ whose integral is zero. Which additional hypothesis, together with continuity, would allow us to conclude that a function with zero integral is identically zero?

One minute, spoken aloud. Explain on the figure why "zero integral" and "zero area" do not mean the same thing.

Delayed review · S79. Calculate an area between trigonometric curves, still looking for the intersections and the sign.

Investigation

Bounding an integral with rectangles - 35 min.

We want to bound $I = \int_0^1 x^2 \,\mathrm{d}x$. For an integer $n \ge 1$, we split $[0;1]$ in n intervals of the same length. Let L_n be the sum of the areas of the rectangles whose heights are the values of x^2 at left endpoints, and U_n the one obtained at the right-hand endpoints.

1. For $n = 4$, draw both families of rectangles, then calculate L_4 and U_4.

2. Justify that $L_4 \le I \le U_4$.

3. Write a Python function with argument n which returns L_n and U_n. Determine algebraically $U_n - L_n$.
4. Determine the smallest integer n for which these bounds are at most 0.01 apart.
5. Calculate the exact value of I using an antiderivative and check that it lies within the bounds obtained for $n = 4$.